01 **Find the derivative of**

$$h(x) = x\sec^{-1}(3x)$$

02 **Find the derivative of**

$$f(x) = x\sqrt{1-x^2} + \cos^{-1}(x^3)$$

03 **Find an equation for the tangent line to the graph of**

$$y = \tan^{-1} x \quad \text{at} \quad (1, \pi/4)$$

04 **Find the inverse-function derivative**

$$(f^{-1})'(2) \quad \text{if} \quad f(x) = x^3 + 2x - 1$$

05 **Find the inverse-function derivative**

$$(f^{-1})'(6) \quad \text{if} \quad f(x) = x^3 - \frac{2}{3}$$

06 **Find the derivative of y with respect to the appropriate variable.**

$$y = \cos^{-1}(x^2)$$

07 **Find the derivative of y with respect to the appropriate variable.**

$$y = \sin^{-1}\left(\frac{3}{t^2}\right)$$

08 **A particle moves along the x-axis, with t ≥ 0. Find its velocity at t = 1.**

$$x(t) = \tan^{-1}(t^2), \quad t = 1$$

09 **Find the derivative of y with respect to the appropriate variable.**

$$y = \sec^{-1}(2s + 1)$$

10 **Find the derivative of y with respect to the appropriate variable.**

$$y = \csc^{-1}(x^2 + 1), \quad x > 0$$

11 **Find the derivative of y with respect to the appropriate variable.**

$$y = \sqrt{s^2 - 1} - \sec^{-1}(s)$$

12 **Find an equation for the tangent line to the graph at x = 2.**

$$y = \sec^{-1} x, \quad x = 2$$

13 **Find an equation for the tangent line to the graph at x = 3.**

$$y = \sin^{-1}\left(\frac{x}{4}\right), \quad x = 3$$